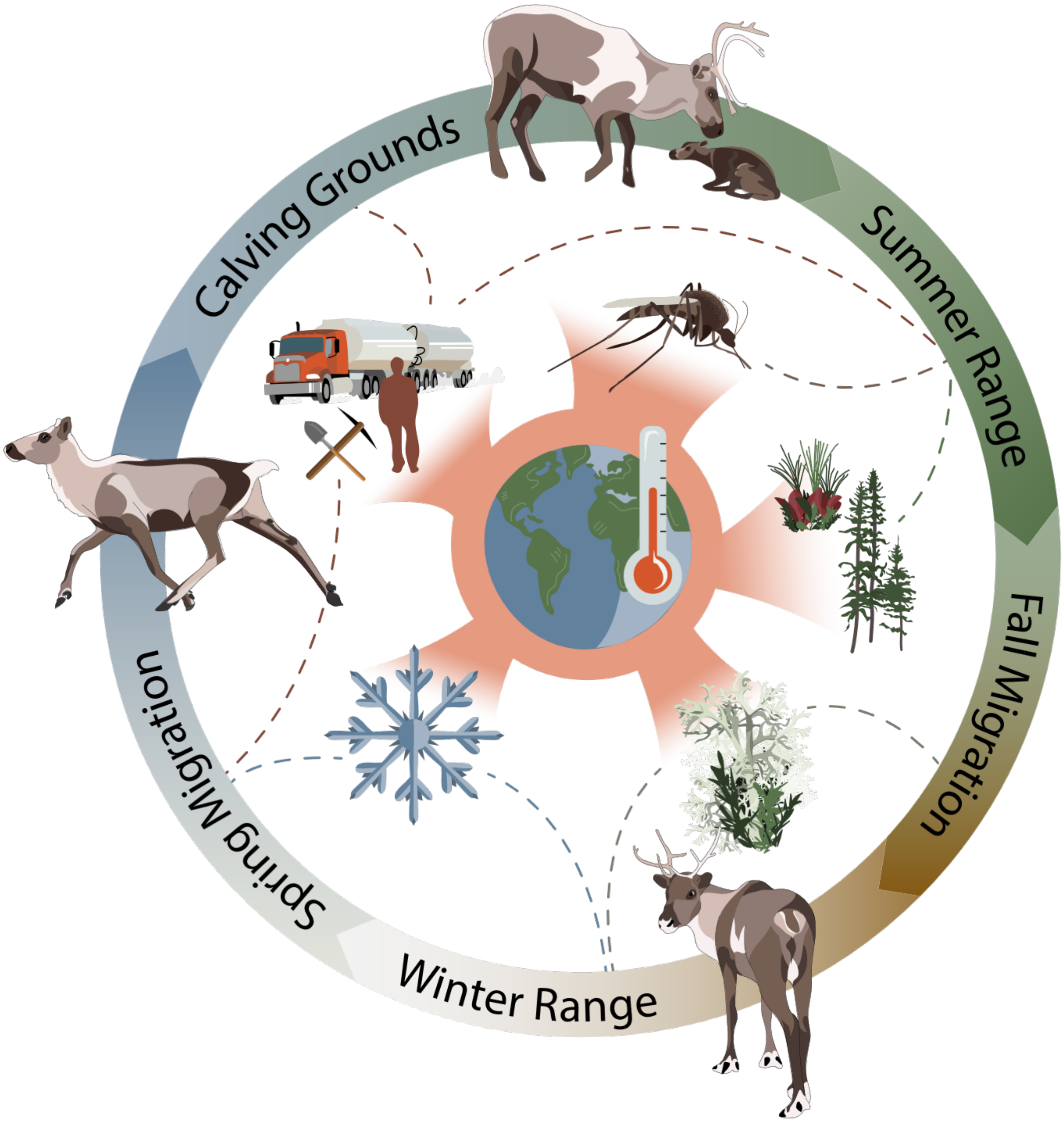
Fate of the Caribou

Research Summary
November 2021 - November 2025
fateofthecaribou.esf.edu

Introduction

The ebb and flow of caribou migrations have shaped the landscape and the lifeways of northern peoples for millennia. As the most widespread and abundant large mammal in the Arctic, and as the world's longest over-land migrator, caribou have a profound impact on their world: they are hunted for meat and materials; they enrich the soil when they defecate or die; they influence vegetation growth by browsing some plants more than others. As human development spreads and the climate warms, caribou migration routes and ranges are changing. While caribou populations fluctuate naturally, some herds have experienced catastrophic population crashes in recent decades, thought to be caused by a combination of climate change-related factors and human influence. Our research seeks to understand how caribou, people, and climate change interact and to help land managers and caribou stewards ensure a future for caribou in Canada and Alaska.

The Fate of the Caribou project is a group of interdisciplinary researchers working to investigate changes in caribou populations and movements, grounded in concerns and observations from Indigenous Elders, hunters, and Knowledge Holders. With experts on caribou population cycles, migration, soundscapes (what caribou hear), and vegetation mapping, we use science to further the collective understanding of shifts in caribou movements, their survival, and how they use their landscapes.



Collectively, our group has applied our ecological knowledge and statistical, mathematical, remote-sensing, and machine learning skills to investigate relationships between caribou and many of the inter-connected factors influencing them: industrial and human development, harassment by biting insects like mosquitoes, changes in the distribution and abundance of vegetation, availability of lichen (caribou's favorite food), snow and ice conditions, and, underlying it all, climate change. Along the way, we have developed new methods for collecting, analyzing, and interpreting data on caribou and their environment and have produced comprehensive datasets and analysis tools so other researchers and land stewards can build on what we have learned.

As the first phase of funding for the Fate of the Caribou Project concludes, we share here our most significant findings. *Masì cho*, *quyana*, *merci*, *quyanainni*, and thank you to the many partners who made this work possible and shared their knowledge with us, including: Wek'èezhìı Renewable Resources Board, North Slave Métis Alliance, Tłı̨chǫ Government, Western Arctic Caribou Herd Working Group, Kugluktuk Angoniatit Association, Government of Northwest Territories, Government of Nunavut, National Park Service, Yellowknives Dene, and the Global Initiative on Ungulate Migration. We look forward to continuing to work with you to explore the relationships between caribou, people, and the land.

The Fate of the Caribou team

[Visit our project website]

Citation

Brose, A., N. Barbour, C. Beaupre, A. Gunn, B. Larue, Q. Liao, M. McConnell, K. Orndahl, M. Perra, S. Salganek, L. Berner, B. Fagan, S. Goetz, E. Gurarie, and M. Hebblewhite. 2025. Fate of the Caribou: Research Summary, November 2021 – November 2025.

Contents

Fate of the Caribou team

State University of New York College of Environmental Science and Forestry

Elie Gurarie – *Primary Investigator*

Ophelie Couriot – *Post-doctoral Researcher*

Nicki Barbour – *Post-doctoral Researcher*

Chloe Beaupre – *PhD Candidate*

Megan Perra – *PhD Candidate*

Sagnik Nandy – *PhD Student*

Anna Brose – *Program & Communications Manager*

Northern Arizona University

Scott Goetz – *Primary Investigator*

Logan Berner – *Primary Investigator*

Katie Orndahl – *Post-doctoral Researcher*

Skye Salganek – *PhD Candidate*

University of Maryland

Bill Fagan – *Primary Investigator*

Marron McConnell – *PhD Candidate*

Qianru Liao – *PhD Candidate*

University of Montana

Mark Hebblewhite – *Primary Investigator*

Benajmin Larue – *Post-doctoral Researcher*

CircumArctic Rangifer Monitoring and Assessment Network

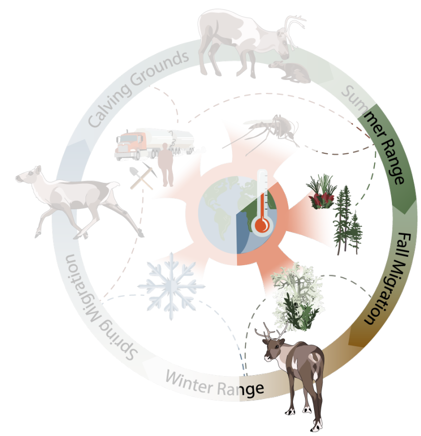
Anne Gunn – *Primary Investigator*

Funding

The Fate of the Caribou Project was funded in part by National Science Foundation Navigating the New Arctic Award #2127271. We are also grateful for financial support from Government of the Northwest Territories Cumulative Impacts Monitoring Program, Environment and Climate Change Canada, and Government of Yukon.

Vegetation & Lichen

Summer is a critical foraging period for caribou populations as they have a relatively brief period to exploit high-quality vegetation needed to recover from the nutritional stresses of winter, migration, and calving and to build fat reserves and body condition in anticipation of the subsequent rut, fall migration, and winter. The overall trend in the Arctic is of rapid warming causing many tundra ecosystems to become more productive, shrubbier, and fire-prone over recent decades, at the expense of non-vascular plants like lichen. We investigated patterns in plant community shifts and how these changes are impacting caribou – and vis versa.



The Arctic Plant Aboveground Biomass Synthesis Dataset

Collecting field data in the Arctic is hard. Field sites are often very remote, with high costs and difficult logistics necessary to access them. The weather can be challenging and harsh. Plus, there are mosquitoes to contend with. In the Arctic Plant Aboveground Biomass Synthesis Dataset, we've pulled together a team of ecologists with, collectively, centuries of experience on the ground across Arctic landscapes. Together, we've aggregated and harmonized years of plant biomass data we've all collected into one easy-to-use database. Aboveground plant biomass is the amount of plant material above the soil (so, not including roots), and is an important factor for, among other things, knowing how much food is available to caribou and other animals. Plant biomass data is broken down into plant types: grasses, forbs (herbaceous plants like wildflowers), shrubs, and trees. Our team has already leveraged this dataset to create maps of vegetation biomass across the entire Arctic biome, but the potential uses for this dataset are many. We hope this dataset can be helpful for researchers, land managers, and communities across the Arctic as we all try to better understand Arctic ecosystems and how they are changing.

Research lead: Dr. Katie Orndahl, katieorndahl@gmail.com

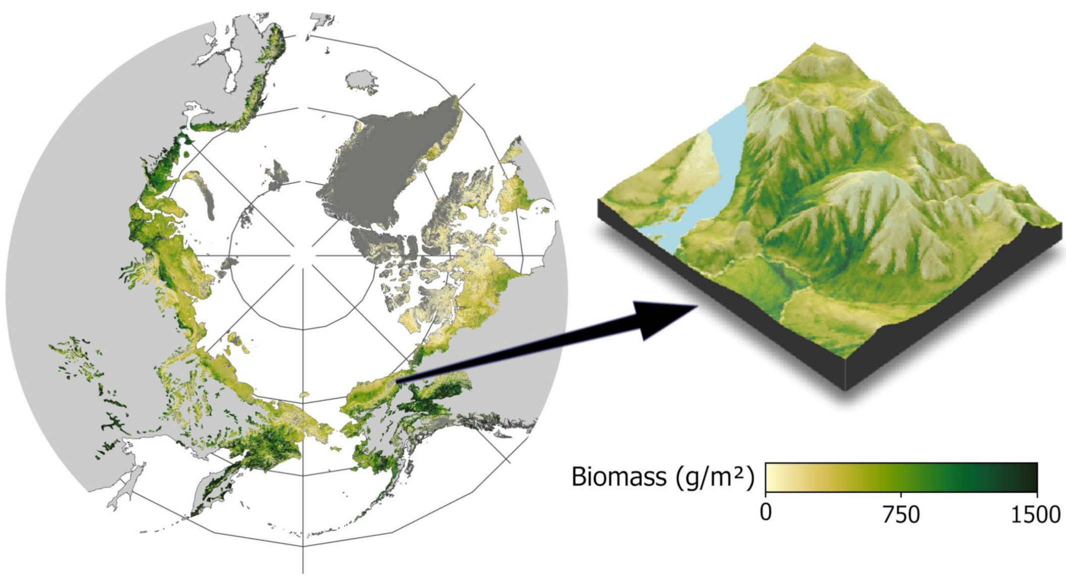
Arctic plant aboveground biomass maps

Maps are an important tool for understanding Arctic landscapes, as they provide seamless information across a broad area. All mapping projects start with high quality field data. This data provides a critical level of detail and accuracy, but because we can't measure everything, everywhere, all at once, field data often amount to a small smattering of points across a large landscape. To complement and stretch these small points, we used computer models and data from The Arctic Plant Aboveground Biomass Synthesis Dataset to infer and map plant and woody plant (shrubs + trees) biomass across the entire Arctic. Measuring plant biomass (i.e., the total weight of all plant species) is important because it captures the full three-dimensional view of Arctic vegetation, not just the percentage of the ground covered in plants, which we can measure from satellite imagery. Furthermore, biomass is sensitive to changes in climate and closely tied to food webs and other ecological functions. Our maps are the highest-resolution maps of vegetation biomass ever produced across the Pan-Arctic – over 70,000 times higher resolution than previous products. These maps provide a complete view of conditions across the entire Arctic region, while still being detailed enough to capture how vegetation changes as you move from a dry, rocky ridge top down to a moist river valley, or across a patterned landscape influenced by permafrost.

Research lead: Dr. Katie Orndahl, katieorndahl@gmail.com

View an interactive map of plant biomass

Access the 30-m resolution estimates of plant biomass



Caribou impacts on vegetation

As a result of climate change, scientists, residents, and land managers alike have noted significant changes in vegetation across the Arctic. We understand that these changes will have consequences. In particular, vegetation changes will alter habitat and forage availability for wildlife species such as caribou. However, we don't often consider how changes in wildlife might impact vegetation. Take, for example, the Fortymile Caribou Herd in Alaska and Canada, which experienced a period of rapid population growth followed by a steep decline. We examined trends in the herd's population, along with trends in vegetation, to understand how the two were related. We found that changes in vegetation were most dramatic in areas with the highest caribou density (caribou per square kilometer), which suggests caribou may be impacting their surroundings. For example, in areas with high caribou density lichen declines were severe; unpalatable deciduous shrubs (like birch and alder) expanded, while palatable shrubs (such as willow) did not. Although we cannot link these vegetation changes directly to caribou, this suggests that across important habitat areas, forage declined because so many caribou were eating it. This may have influenced other changes observed in the herd, such as demographic trends, range expansion, and the population's eventual decline.

Research lead: Dr. Katie Orndahl, katieorndahl@gmail.com

Changes in plant aboveground biomass

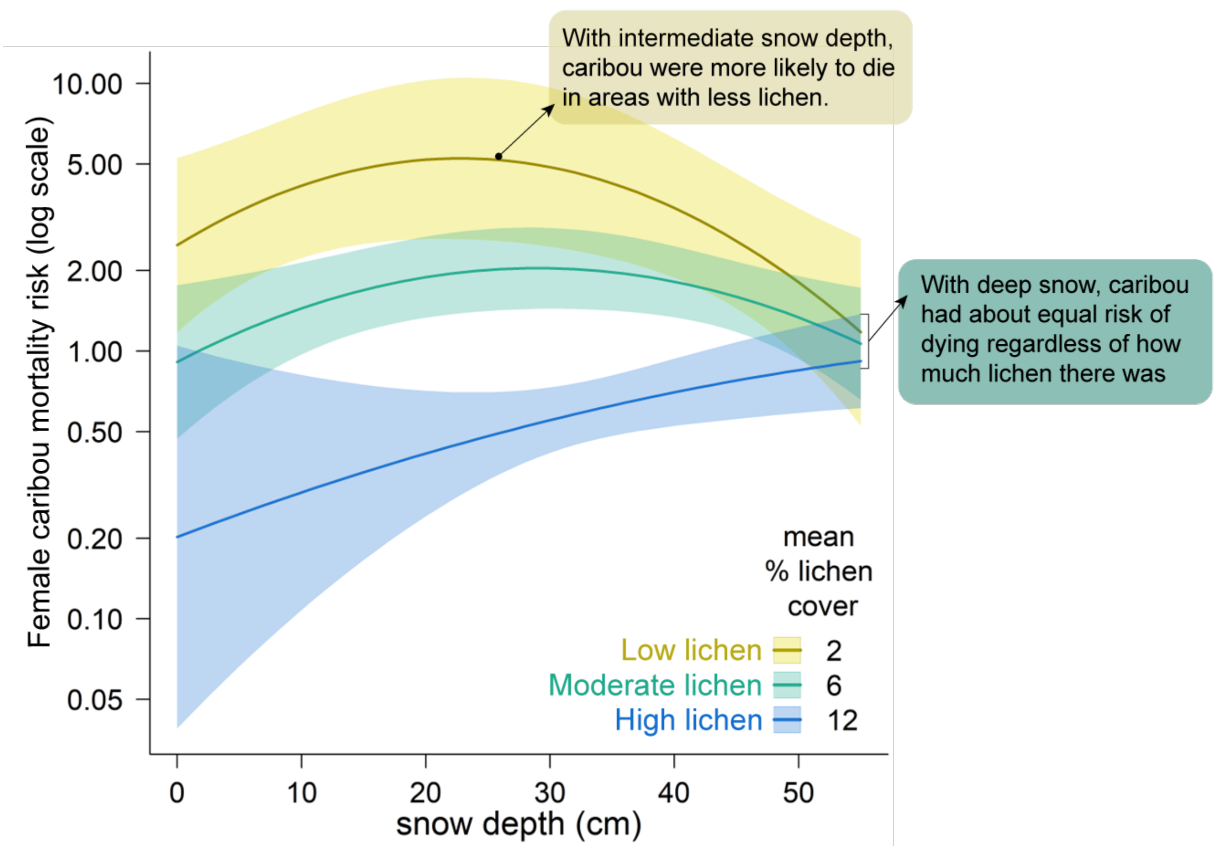
The Arctic is warming faster than anywhere else on Earth, placing tundra ecosystems at the frontlines of climate change. Vegetation, in particular, is changing rapidly. We used our new, high-resolution maps of plant and woody plant biomass (i.e. the total weight of all plant species) across the Arctic and used them to understand how vegetation is changing.

From 2000-2020, plant biomass increased ~14%. Woody plants led the charge, suggesting a warmer Arctic will be a shrubbier Arctic – which has major implications for species like barren-ground caribou, who rely on open tundra. Importantly, these changes varied across space leading to a patchwork of 'greening' and 'browning.' Understanding what causes this patchwork is of crucial importance. Our maps capture patterns of vegetation change driven by past disturbance (e.g., wildfire) and environmental conditions (e.g., amount of rainfall). Increasing vegetation (greening) was tied to warmer temperatures, but also areas with more snow cover. This highlights the importance of winter conditions for Arctic vegetation change: plants thrive when they are well-insulated by snow during the harsh winter. These maps of vegetation changes are freely available and will help land managers and biologists better understand how climate change is affecting plants and the animals that eat plants.

Research lead: Dr. Katie Orndahl, katieorndahl@gmail.com

Lichen & snow depth

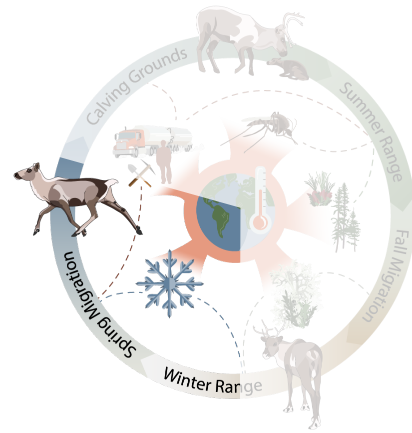
We examined how lichen abundance and snow depth interact to affect overwinter survival in Western Arctic Herd caribou. We found that lichen-rich habitats reduce mortality risk up to 9-fold compared to low-lichen areas, but the benefits of lichen diminish when deep snow limits forage access. These findings suggest that Arctic-wide lichen declines could threaten caribou populations by reducing overwinter survival, particularly as the benefit of lichen-rich habitat is critically dependent on snow conditions remaining moderate enough for caribou to access their food.



Research lead: Chloe Beaupré, cbeaupre@esf.edu

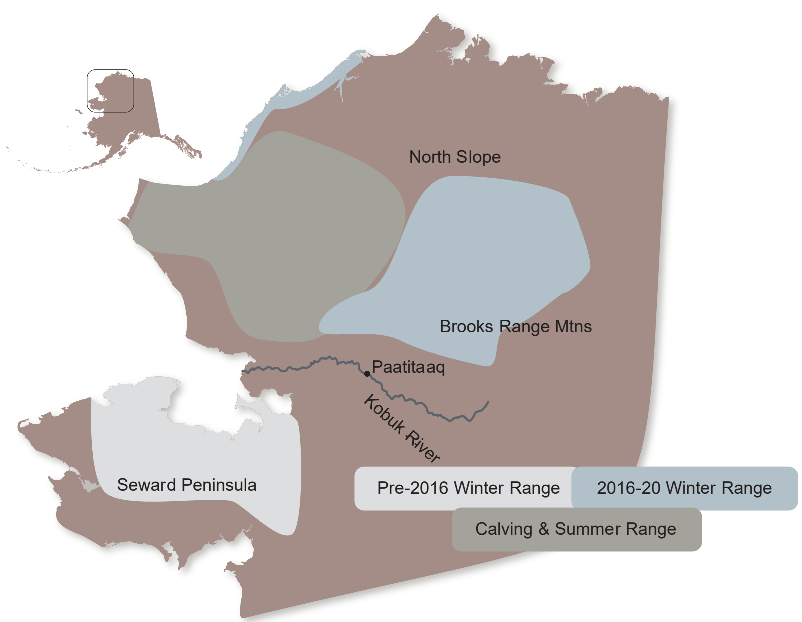
Snow & Ice

Snow covers the Arctic and boreal regions for approximately 9 months per year, with tremendous impacts on the morphology and functioning of Arctic ecosystems [98]. Snow presence and snowmelt timing appear to have little impact on caribou [34,98,99], but multiple studies indicate that the quality of snow, in particular its depth and softness, can be important for access to food and mobility. We examined ways snow, ice, rain-on-snow events, and freeze-thaw timing of lakes and rivers impacted caribou survival and movement patterns.



Winter conditions and Western Arctic Herd survival

Caribou in the Western Arctic Herd depend on ground lichens as their primary winter forage, but increasingly-variable snow and ice conditions are limiting access to this resource. Warmer, wetter winters create cycles of melting and refreezing and rain-on-snow events, which form ice layers within the snowpack. These ice crusts block access to lichens and increase the energetic costs of foraging during a critical time of year. Using GPS tracking collar data from 199 adult female caribou over 14 winters, we found that caribou generally selected areas with higher lichen cover but avoided sites with deep snow and ice. The herd's recent shift away from its traditional southern winter ranges occurred during several consecutive years of severe icing, suggesting that these events were a key driver of the change in the herd's winter range. During those years, Western Arctic herd caribou favored shorter migrations and mountainous areas where icing is less common, highlighting how climate-driven snow and ice dynamics are reshaping winter habitat use and migration strategies.

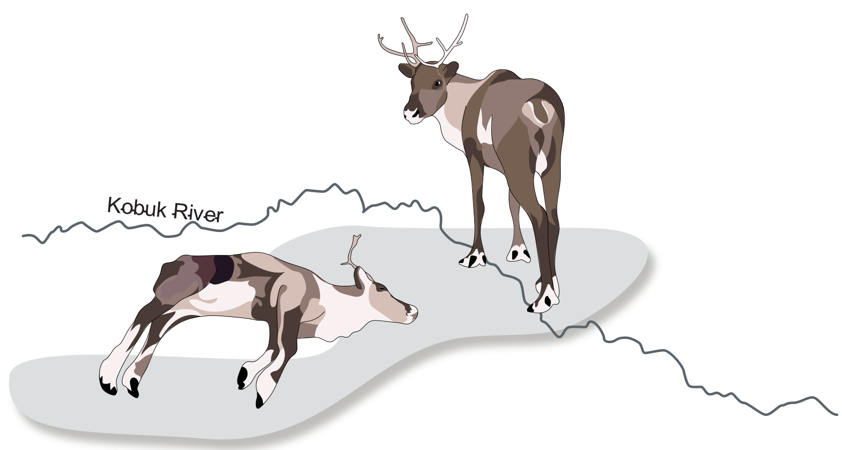


Research lead: Skye Salganek, dss66@nau.edu

Social memory affects winter range choices

Between 2017 and 2022, the Western Arctic Caribou Herd in Alaska largely shifted its winter range from the Seward Peninsula to the Brooks Range mountains and North Slope. Their use of traditional river crossings of the Kobuk River, such as the portage at Paatitaaq (Onion Portage), have diminished. We analysed the likelihood of caribou surviving if they stayed north of the Kobuk River, wintering in the Brooks Range or North Slope, compared to their changes of surviving if they wintered south of the Kobuk River, on the Seward Peninsula. We found that survival was not better for caribou who wintered north of the Kobuk River, but it *was* more predictable. Survival south of the Kobuk River could be high, but only when environmental conditions were ideal, suggesting that shifts in the Western Arctic Herd's winter range may have been an adaptation to unpredictable winter conditions on the Seward Peninsula.

The best predictor for whether caribou decided to migrate south of the Kobuk River in a given year was how high survival was south of the river in the previous year. In other words, caribou remembered if many of their companions died the previous winter and, if so, were more likely to winter north of the Kobuk River. This is evidence – long reflected in Alaska Native knowledge of caribou behavior – that these large-scale movements are informed by the herd's collective social memory.



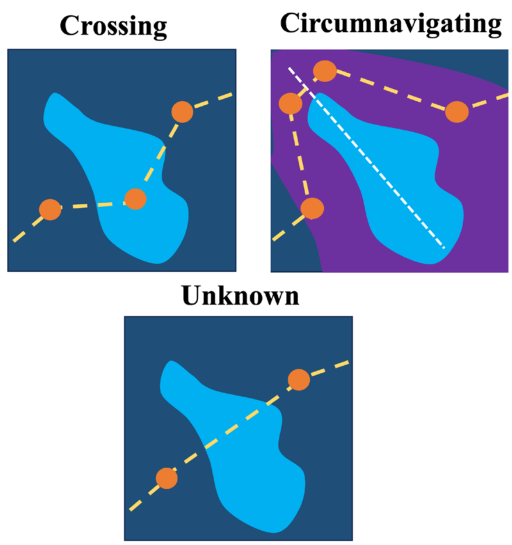
Since this study was published, some caribou have begun crossing the Kobuk River to winter near the Seward Peninsula again. Perhaps they've decided to try their luck with the unpredictable, but potentially ideal winter conditions, or decided the mediocre survival in the Brooks Range wasn't their best chance. Time will tell whether their gambit succeeds.

Research lead: Dr. Elie Gurarie, egurarie@esf.edu

Read the scientific paper

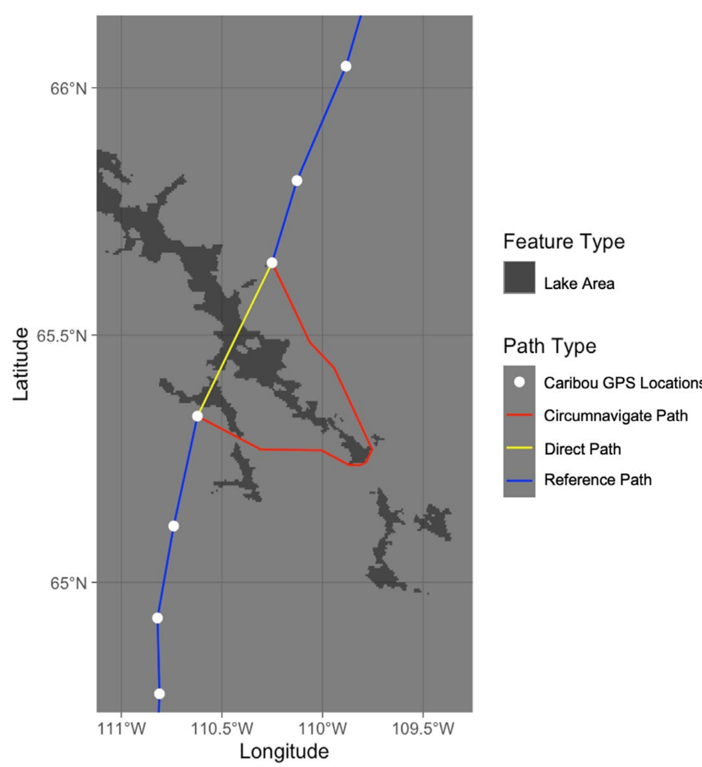
View a graphic research summary

Lake melting & freezing affects migration decisions



How do the timing and patterns of lake ice freezing and thawing shape migratory behavior in Arctic caribou? Rising Arctic temperatures are causing earlier ice melt and later freeze-up, disrupting frozen corridors that caribou rely on for efficient travel to calving grounds. These changes delay migration, raise energetic costs, and threaten the survival of females and calves. With the Bathurst caribou herd population collapsing from 480,000 to fewer than 7,000, understanding these disruptions is urgent.

We analyzed 20 years (2001-2021) of GPS tracking collar data for 406 adult caribou and daily MODIS measurements of surface reflectivity (how bright the surface appears to satellites, which indicates snow and ice cover) at Contwoyto Lake, a large (>100 km) lake in northern Canada, to link fine-scale lake ice dynamics with seasonal water-crossing behaviors by caribou. In other words, we examined whether ice conditions on Contwoyto Lake influenced whether caribou crossed directly over the lake or went around it. Our models revealed distinct seasonal drivers of crossing behaviors: in spring, caribou were more likely to cross when surface reflectivity was relatively high, at or above the 56th percentile of yearly values, indicating strong ice. By contrast, in autumn, when lakes were ice-free, crossing decisions were better explained by movement-related factors, such as the speed the caribou were traveling.

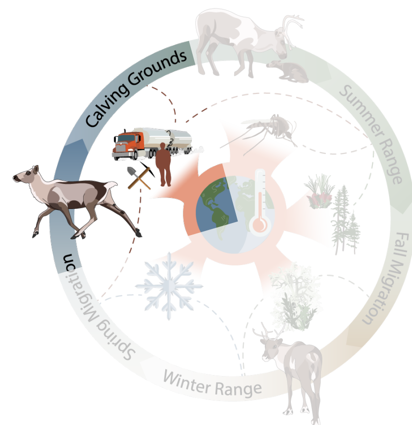


These findings show that ice acts as a seasonal filter, shaping migration through both the perception of safe routes and the energetic cost of travel. By linking satellite data to individual animal movement, our framework provides a transferable tool for detecting early-warning signals of climate-driven migration disruption in rapidly-changing ecosystems.

Research lead: Qianru Liao, qianru@umd.edu

Read the scientific paper

Habitat Disturbance & Industry

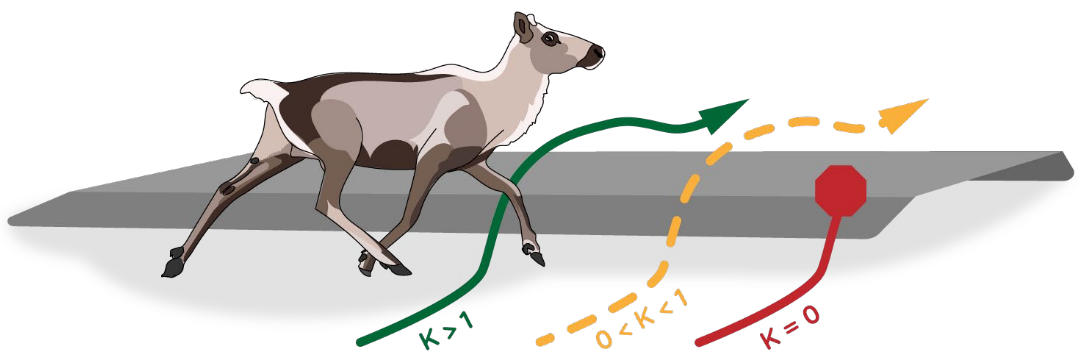


Human land use is the main source of global change contributing to biodiversity loss around the world. Land use change can lead to direct habitat loss and fragmentation through resource extraction, energy development, and transportation network development, as well as indirect habitat loss through behavioral avoidance of areas surrounding human disturbance. The most common forms of disturbance in barren-ground caribou ranges are mining, oil and gas installations and explorations, and roads. Independent studies have shown that caribou avoid these disturbances at large and variable spatiotemporal scales. We built tools and ran analyses to quantify the behavioral response of caribou to roads; these tools are already being used by caribou researchers to understand the potential impacts of proposed roads in several caribou herd ranges.

The Permeability R Package

Caribou are sometimes reluctant to cross roads, power lines, rivers, etc. ("linear barriers"), even during critical migration periods. They are more willing to cross some barriers than others, making those barriers more "permeable". The willingness and ability of caribou and other species to cross linear barriers, both human-made and natural, is important for conservation practitioners and land stewards. We therefore developed a straightforward method to estimate the permeability of linear barriers to animal movement, available in the computer program "R" package *permeability*.

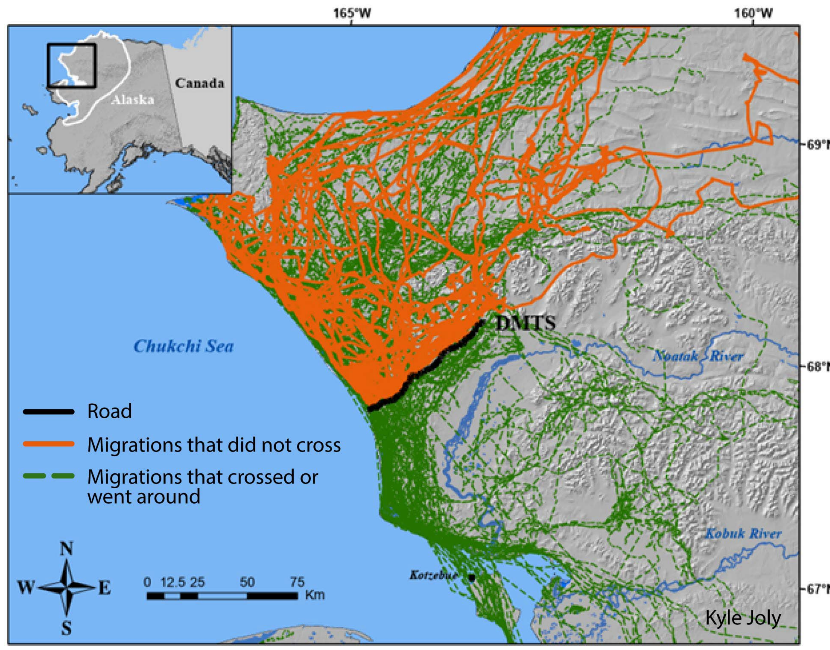
The package was developed specifically to model the permeability of roads to barren-ground and boreal caribou in the Northwest Territories, Canada, where proposed development of new, all-season roads would bisect caribou ranges, and where caribou already have extremely low crossing rates of existing seasonal roads. However, the tool is applicable to any animal movement dataset that interacts with linear features. The *permeability* package allows for permeability of a linear barrier to be quantified as an index, κ, that ranges from total impermeability (κ = 0, no crossings), intermediate permeability (0 < κ < 1, some crossings possible but less than if there were no barrier), or hyper-permeability (κ > 1, crossings more likely than if there were no barrier). Using "maximum likelihood" statistical techniques, the *permeability* package can incorporate covariates (e.g., sex of animal, time of year, vehicle traffic, barrier structure) to model permeability under different conditions, making it a useful tool for land stewards operating in complex real-life systems. We have applied the *permeability* tool to both boreal and barren-ground caribou datasets in the Northwest Territories, with results showing low permeability of high-traffic roads.



Research lead: Dr. Nicki Barbour, nbarbour@towson.edu

Download the permeability package

Barriers impact caribou survival



Roads and large-scale infrastructure can affect the movements of caribou causing general avoidance, delays in crossing, altered behaviors and, in some cases, can prevent caribou from reaching their wintering grounds. In northwest Alaska, a solitary, 80-km long industrial road connecting a large zinc and lead mine to a port affects migrations of Western Arctic Herd caribou. We used GPS collar tracking data to assess whether caribou whose autumn migrations were altered by the road experienced higher mortality risk compared to those whose movements were unaltered. Of the tracked migrations, 101 came within 20 km of the road; 58% of those displayed altered movements.

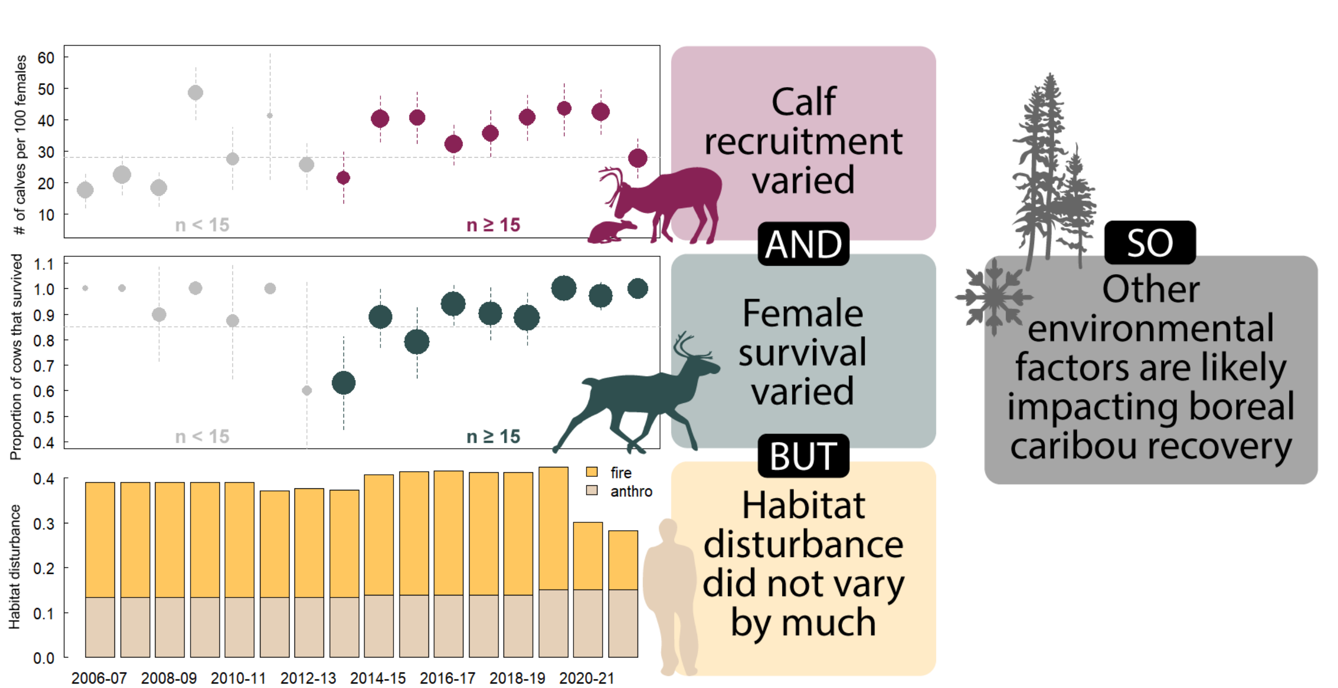
Caribou whose movements were unaltered by the road were 20% more likely to survive until the next spring than caribou whose movements were altered by it, though the results of this statistical test were not strong enough to prove this beyond doubt. Caribou that did not get past the road at all (i.e., the road acted as an impermeable barrier and the caribou spent the winter north of it) were much less likely to survive until the calving period the following spring than caribou who did cross or circumnavigate the road. Of the caribou who were stuck north of the road, 58% survived until calving; of the caribou who went across or around the road, 78.5% survived until calving. This is likely because caribou stuck north of the road for the winter had less access to food, being further away from their lichen-rich wintering areas, and/or because they wasted energy (and, therefore, precious fat reserves) trying to get around the road. We estimate the absolute number of additional mortalities associated with road impermeability to be >1,100 caribou/year, on average. Given our results, we posit that enhancing the permeability of roads by limiting traffic levels, road dust, and noise, among other things, could improve the survival of caribou in the region. *Note: as of November 2025 the peer-reviewed paper for this study (Joly et al.) is in review at Scientific Reports.*

Research lead: Dr. Elie Gurarie, egurarie@esf.edu

Boreal caribou recovery & habitat

The Canadian boreal caribou national Recovery Strategy aims to maintain at least 65% undisturbed habitat within each herd's range to support self-sustaining populations. "Habitat disturbance" includes human disturbance (converting habitat to industry or development) and wildfires up to 40 years old. Using data on caribou survival and caribou calf recruitment (calves that survive to adulthood) from seven monitoring areas in southern Northwest Territories (NWT), we evaluated the Recovery Strategy framework, alternative metrics of habitat disturbance, and weather variables. Adult female caribou survival and caribou calf recruitment are two important measures of a population's ability to grow.

Preliminary results indicate the Recovery Strategy models were poor predictors of boreal caribou survival in the NWT. Human disturbance had little effect on adult female caribou survival, while the impact of fire varied with the number of years since the fire; combining all fires ≤40 years old did not adequately track relationships with adult caribou survival or calf



recruitment. Calf recruitment was negatively associated with linear feature density and 0-year fires, whereas younger burns (1–10 years) were positively associated with survival. In other words, calf recruitment was lower in areas with many "linear features" (roads, seismic lines, etc.) or areas that burned that year, and calf recruitment was higher in areas where there were wildfires in the previous 10 years. Weather, particularly the previous year's snow depth, was a strong predictor, negatively impacting adult female caribou survival, highlighting the influence of environmental conditions, not just habitat disturbance, as important recovery factors.

Current Canadian Recovery Strategy metrics may not fully capture the factors driving NWT caribou demography: Wildfire burn age and environmental conditions both influence caribou calf recruitment and adult female caribou survival. Assessment and management of habitat disturbance should incorporate more specific wildfire burn age and weather conditions into habitat thresholds.

Research lead: Chloe Beaupré, cbeaupre@esf.edu

Calving & Calving Grounds

Caribou are at their most vulnerable as newborn calves and when tending to newborn calves. Furthermore, environmental conditions and forage availability during early life of calves can have lifelong impacts on a calf's ability to survive and reproduce. Caribou herds show high fidelity to their calving grounds, returning generation after generation to areas where they can maximize forage availability while minimizing the threat of predation. The calving period and calving grounds are, therefore, extremely important to the health of caribou populations, and are the focus of much research, policy debate, and concern for caribou stewards.

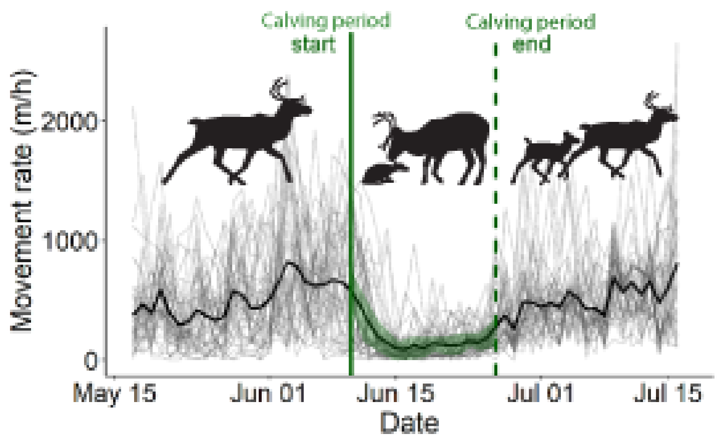
Grizzlies on the calving grounds

Working with northern communities, we are studying interactions between caribou and predators on the Bathurst caribou herd's calving grounds using a large grid of 96 motion-triggered cameras. The cameras take a picture each time something moves in front of them, allowing us to tally the number of times each species was "detected". This community-led project, developed in partnership with the Kugluktuk Angoniatit Association and the Government of Nunavut, provides a non-invasive way to monitor caribou, their calves, and predators during the calving season. The cameras were deployed by community members across 1,600 km2 of the calving grounds before the calving period. Our initial analyses revealed that grizzly bears were detected on the calving grounds five times as much as wolves, suggesting bears may play a larger role in limiting calf survival than previously recognized. We also showed that the peak calving date of the herd can be accurately estimated from camera detections, offering an efficient, scalable, and non-invasive monitoring tool. We are now working on revisions of a peer-reviewed paper outlining our methods and results and developing an infographic of the results that will be translated into Inuinnaqtun so that findings are accessible to Kugluktuk community members.

Research lead: Dr. Benjamin Larue, benjaminlarue5@hotmail.com

What is a calving ground?

Land use planning projects, industrial development proposals, and environmental conservation policies often seek to map sensitive caribou calving areas. However, historic inconsistencies and disagreements about which areas are defined as "calving grounds" have convoluted conservation efforts. We developed a robust method to identify a herd's core calving grounds, and important areas around the core that are used for calving and calf rearing (feeding and caring for young calves before they are fully mobile). Unlike many other calving ground estimates, our methods use both space-use (i.e., what percentage of tracked caribou are within an area during the calving period) *and* behavioral cues that indicate calving to identify the most critical calving areas.

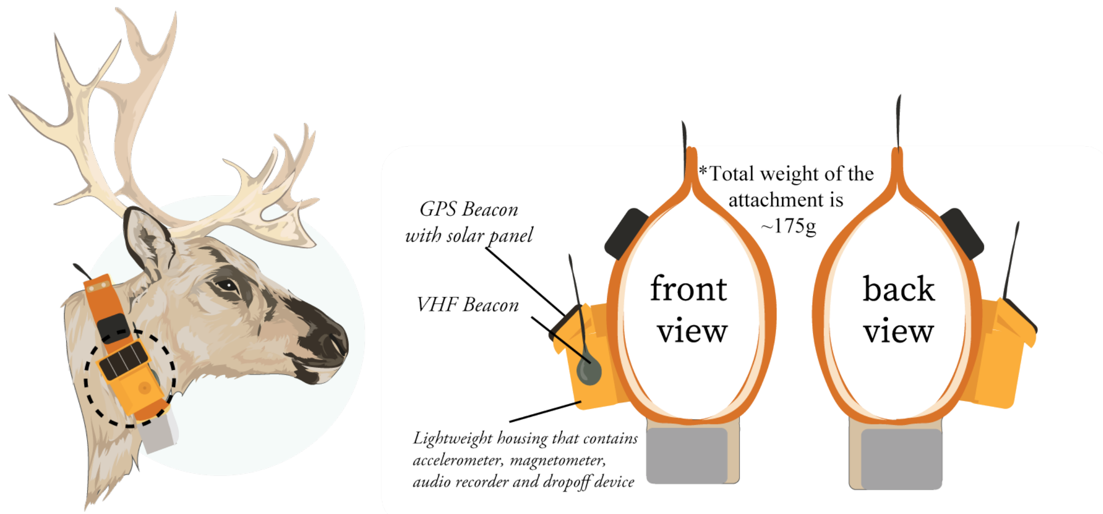


Using GPS tracking data, we can very closely estimate when a caribou gives birth, because her movement suddenly slows. We can also detect when she and her calf are in the early-rearing stage, because her movement starts again but is slow. Then, we can detect when the calf is fully mobile and the pair can move freely because the mother's movement rates return to what they were before the calf was born. By detecting calving events and the subsequent early-rearing period, we can better delineate the core calving grounds and areas important for young calves. Our model provides results that are comparable across years: we used this method to document consistent shifts in the Qamanirjuaq Caribou Herd's calving grounds. This analytical tool leverages large GPS datasets to provide consistent, quantitative, and reproduceable estimates of calving grounds, which is critical to effective conservation strategies and understanding wide-ranging, social species like caribou.

Research lead: Dr. Ophélie Courtiot, ocouriot@alaska.edu

Detecting calving & behavior through sound

Getting fine-scale, on-the-ground data on caribou behavior and life events is difficult, costly, and can be disruptive to caribou. Environmental acoustic monitoring – using weatherproof recording devices to record animals – is a growing field thanks to recent improvements in battery technology and machine learning, which can process thousands of hours of acoustic data. We designed and developed novel acoustic recorders that have been mounted to GPS tracking collars to measure the soundscape caribou experience. This involved building and programming drop-off devices (mechanisms that allow the recording device to fall off the GPS collar after a set amount of time) and acoustic recording devices that have enough memory but a small enough battery to collect a sufficient sample size while not being cumbersome to the animal.



The devices were deployed on 30 free-ranging caribou in partnership with Government of Northwest Territories in spring 2025, and the majority have been successfully recovered. From the audio data, we were able to identify calving events and cow-calf interactions, as well as fine-scale foraging patterns and rumination. We have not yet identified any calf mortalities but are confident this tool could be used to identify calf mortalities by tracking cow-calf interactions through time. We are also using the acoustic data to detect harassment by biting insects, infestation by parasites, exposure to traffic and aviation noise, and other life events.

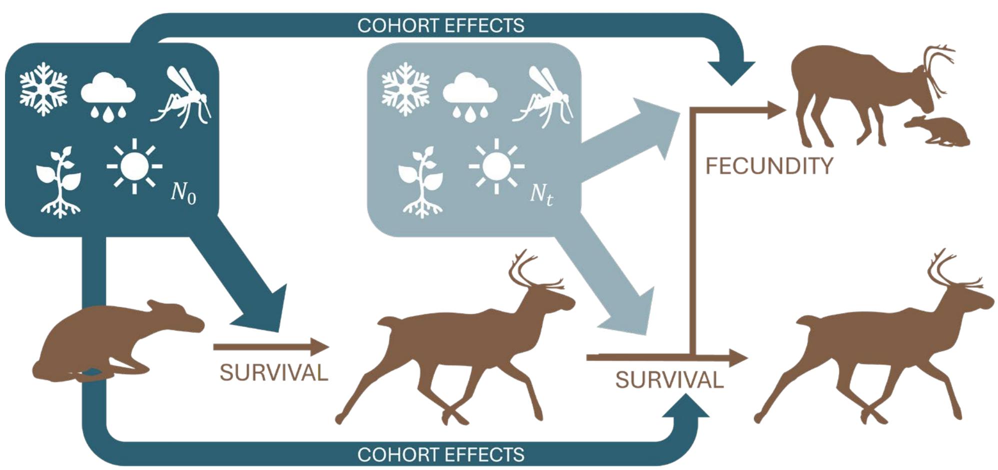
These collars will allow researchers to gather extensive data on the daily experiences and behaviors of large ungulates. While collars have been used for several decades to track animal movements across the landscape, pairing that movement data with acoustic information will provide unprecedented and fine-scale insight into the drivers behind observed behaviors.

Research lead: Megan Perra, mperra@esf.edu

Visit the Caribou Sound Library

Population modeling

Barren-ground caribou are known to experience long-term and large-scale increases and decreases in numbers ("population cycles"). While regular fluctuations in abundance of small animals are often caused by tightly-coupled predator-prey and food availability dynamics, , population cycles in large animals are less well understood. We developed a series of mathematical models of caribou population dynamics incorporating cohort effects, density dependence and environmental variation. Cohort effects refer to how conditions during animals' birth and early life impact future survival and reproduction (e.g., calves born during a wet spring may have smaller body size and decreased survival over their lifetime than a "cohort" of calves born in a drier spring). Density dependence is a phenomenon where survival and reproduction rates change with population size (e.g., probably of survival for an individual animal is lower when the population is very high and there is less food to go around). Environmental variables are things like winter snow depth, summer temperatures, or even major overarching climate patterns, temperature, or amount of rainfall.

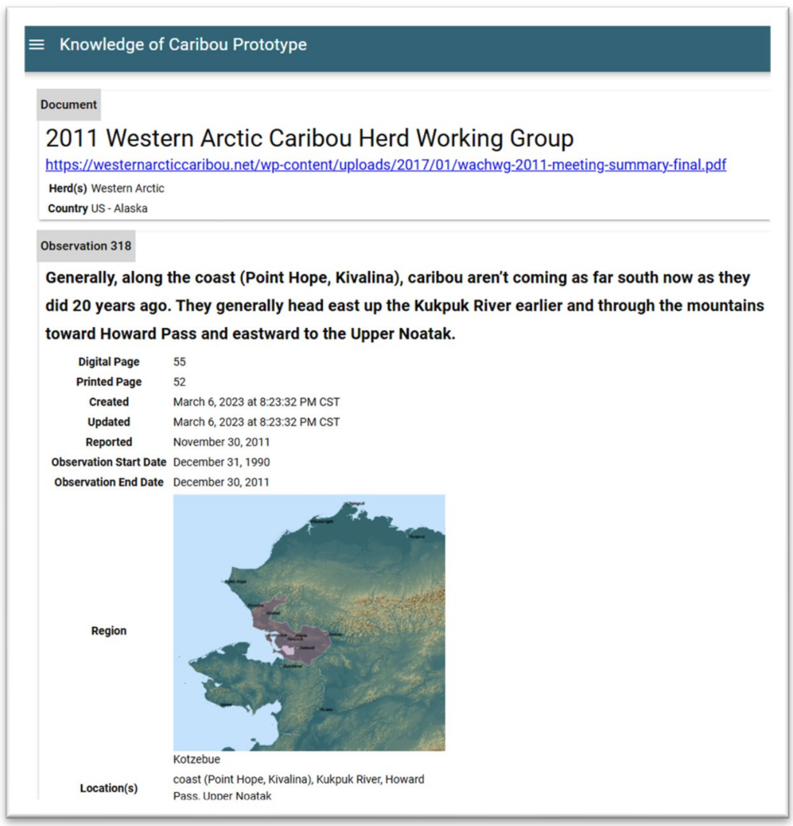


We analyzed the models for their capacity to mimic population cycles and investigated which model components were most crucial in driving these fluctuations. Cohort effects impacting survival were the strongest driver of population cycles across a variety of demographic and environmental scenarios. In other words, conditions during the birth and early life of a cohort of calves influenced that cohort's chances of survival throughout life; in turn, that cohort effect has a strong impact on the ups and downs of the herd's population cycle. Cohort effects related to survival had a greater effect on population cycles than density dependence, environmental variation, or cohort effects related to reproductive success. Our model predictions of population cycle characteristics (such as duration of the cycle and average period) agreed with cycle descriptions documented by Indigenous knowledge holders. This work provides new insight into how complex demographic and environmental interactions drive population cycles in larger and longer-lived animals like caribou.

Research lead: Marron McConnell, mecmcc@umd.edu

Traditional Knowledge database

Local observations and knowledge have been routinely recorded at annual caribou workshops and herd-specific working group meetings throughout the years. Participation in these meetings represents a significant time burden on local community members to attend, often traveling hundreds of miles to do so, and to share their knowledge. These observations provide an important local record of the past state of the Arctic environment and caribou. However, the thousands of pages of meeting notes, while largely publicly available, are not cataloged in such a way that information is searchable. Meanwhile, Traditional Knowledge Holders are frequently approached repeatedly by different researchers seeking knowledge of caribou, requiring them to extend further effort and time to share knowledge they already gave at public meetings.



We built a database prototype, the "Knowledge of the Caribou" database, to catalog Traditional and Local Ecological Knowledge of caribou, pulling observations from public meeting documents and formatting them into a searchable repository. To date, we have entered observations from the Western Arctic Caribou Herd Working Group annual meetings for 2011-2013 (~460 observations) and are working on continuing data entry as we fine-tune the database structure. Database development will continue as funding is available. We hope the Knowledge of the Caribou database becomes a useful tool for caribou Knowledge Holders, caribou co-management and advisory groups, and researchers (in that order).

Research lead: Dr. Ophélie Courtiot, ocouriot@alaska.edu

A Future for Caribou: Perspective from Anne Gunn

The wonder of caribou and their astonishing Arctic migrations across snow-clad tundra has been the heartbeat sustaining and enriching our Fate of the Caribou research journey. It has not always been easy given the realities of Arctic research and that we joined the caribou at a low point in their natural population cycle. As their numbers declined, their seasonal range shrank northward, putting them even further away and stretching our budgets ever more. But the Fate of the Caribou team had the inestimable advantage of meeting and working with people who have long shared the landscape with caribou. Their stories and concerns shaped our knowledge of caribou and encouraged us to reciprocate by using our skills to produce tools that empower local people to help caribou and steward their shared landscape through exceptional changes.

The caribou are already feeling the warming climate, especially changes such as increases in the tundra shrub growth. In particular, caribou are adapted to cold and snow and not heat. As the number of hot days (>19oC) is projected to increase fourfold for the Bathurst herd even under a relatively optimistic climate scenario, caribou will suffer heat stress. Not all caribou herds will be threatened to the same extent and herds along the cooler Arctic coasts are likely, at least initially, to fare better in a warming climate. But without reducing greenhouse gases and slowing the rate of warming, the outlook over the next 100 years, even for the coastal herds, is grim.

When we look back 1000s of years, we know that the hallmark of caribou is their adaptability to the often-rapid and sweeping climate shifts as the glaciers advanced and retreated. The caribou also advanced and retreated and, overall, they survived by having the freedom to select where to go. But these days, the landscape is being etched not by caribou migration trails but by all-season roads, many of which are supply routes for mines and oilfields. We know that managing traffic to create "traffic breaks" allows caribou to cross and continue their seasonal migrations.

The Fate of the Caribou research findings contribute knowledge and tools for caribou recovery, but we need to ensure how we leave a lasting legacy by ensuring our tools and research results are easy to use, applicable, and available. Our research has to return the trust and contributions of the Elders and caribou users who shared their experience and knowledge when we started this journey with the caribou and caribou people.

As if a warming climate and spread of roads were not enough, the caribou are mostly at the low point in their long-term population cycles. The caribou on the landscape now are barely a fraction of what we would have seen 30 years ago for most herds. As caribou numbers dropped, the space they occupied shrank and shifted leaving stretches of the tundra empty, but which will be needed when caribou populations recover. The return of the caribou will depend on collaborative land use and herd management planning, decisive climate action, and interdisciplinary, culturally-informed research. We must do our part; the caribou will do the rest, as they always have.